

$$\begin{aligned} & \frac{1}{m_e^2} S_Y^2 \overline{y_d}^{i2i3} y_d^{i1i4} + \frac{1}{16\pi^2} \left(-\frac{7}{12} \frac{1}{m_e^2} (S_Y^2 (27 \overline{y_d}^{pR} \overline{y_d}^{i2i3} (y_d^{p14} y_d^{i1r} (1 + S_Y^2) - 8 C_Y^2 y_d^{pR} y_d^{i1i4}) + \right. \\ & \quad \overline{y_d}^{i2i3} (2 y_d^{i1i4} (17 g_1^2 + 81 g_2^2 - 84 g_3^2) - 9 y_d^{p14} \overline{y_u}^{pR} y_u^{i1r} (-1 + C_Y^2)) + \\ & \quad \overline{y_d}^{p13} (9 y_d^{i1i4} (3 \overline{y_d}^{i2r} y_d^{pR} (1 + S_Y^2) - \overline{y_u}^{i2r} y_u^{pR} (-1 + C_Y^2)) + 16 g_3^2 y_d^{p14} \delta_{i1i2})) + \\ & \quad 8 g_3^2 (S_Y^2 \overline{y_d}^{i2p} y_d^{i1p} + C_Y^2 \overline{y_u}^{i2p} y_u^{i1p}) \delta_{i3i4} + \frac{2}{3} g_3^4 LF_{3,0}[m_3] \delta_{i1i2} \delta_{i3i4} + \\ & \quad g_3^4 LF_{4,-1}[m_3] \delta_{i1i2} \delta_{i3i4} - \frac{16}{15} g_3^4 LF_{5,-2}[m_3] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{1}{2} \sum_p S_Y g_1^2 \frac{1}{m_e} \overline{y_d}^{i2i3} y_d^{i1i4} (2 S_2 y_C y_Y + S_Y C_2 y_Y) LF_{1,0}[m_d^p] + \\ & \quad \frac{2}{9} \sum_p g_3^4 LF_{3,0}[m_d^p] \delta_{i1i2} \delta_{i3i4} - \frac{5}{12} \sum_p g_3^4 LF_{4,-1}[m_d^p] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{8}{45} \sum_p g_3^4 LF_{5,-2}[m_d^p] \delta_{i1i2} \delta_{i3i4} - 3 S_Y \frac{1}{m_e^4} \overline{y_d}^{pR} \overline{y_d}^{i2i3} y_d^{pR} y_d^{i1i4} (-S_2 y_C y_Y + S_Y^3) LF_{1,0}[m_d^r] - \\ & \quad \frac{1}{2} \sum_p S_Y g_1^2 \frac{1}{m_e^4} \overline{y_d}^{i2i3} y_d^{i1i4} (2 S_2 y_C y_Y + S_Y C_2 y_Y) LF_{1,0}[m_e^p] + \\ & \quad S_Y \frac{1}{m_e^4} \overline{y_d}^{i2i3} y_d^{i1i4} \overline{y_e}^{pR} y_e^{pR} (S_2 y_C y_Y - S_Y^3) LF_{1,0}[m_e^r] + \\ & \quad -\frac{1}{2} S_Y \frac{1}{m_e^4} \overline{y_d}^{i2i3} y_d^{i1i4} (2 \overline{y_e}^{pR} y_e^{pR} (S_2 y_C y_Y - S_Y^3) + \sum_p g_1^2 (2 S_2 y_C y_Y + S_Y C_2 y_Y)) LF_{1,0}[m_l^p] - \\ & \quad \frac{1}{2} S_Y \frac{1}{m_e^4} \overline{y_d}^{i2i3} y_d^{i1i4} (6 \overline{y_d}^{pR} y_d^{pR} (-S_2 y_C y_Y + S_Y^3) + 6 C_Y \overline{y_u}^{pR} y_u^{pR} (S_2 y_Y + S_Y C_Y) + \\ & \quad \sum_p g_1^2 (2 S_2 y_C y_Y + S_Y C_2 y_Y)) LF_{1,0}[m_q^p] + \frac{4}{9} \sum_p g_3^4 LF_{3,0}[m_q^p] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{5}{6} \sum_p g_3^4 LF_{4,-1}[m_q^p] \delta_{i1i2} \delta_{i3i4} + \frac{16}{45} \sum_p g_3^4 LF_{5,-2}[m_q^p] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \sum_p S_Y g_1^2 \frac{1}{m_e^4} \overline{y_d}^{i2i3} y_d^{i1i4} (2 S_2 y_C y_Y + S_Y C_2 y_Y) LF_{1,0}[m_u^p] + \frac{2}{9} \sum_p g_3^4 LF_{3,0}[m_u^p] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{5}{12} \sum_p g_3^4 LF_{4,-1}[m_u^p] \delta_{i1i2} \delta_{i3i4} + \frac{8}{45} \sum_p g_3^4 LF_{5,-2}[m_u^p] \delta_{i1i2} \delta_{i3i4} - \\ & \quad 3 S_Y C_Y \frac{1}{m_e^4} \overline{y_d}^{i2i3} y_d^{i1i4} \overline{y_u}^{pR} y_u^{pR} (S_2 y_Y + S_Y C_Y) LF_{1,0}[m_u^r] - \frac{1}{4} S_Y \frac{1}{m_e^4} \overline{y_d}^{i2i3} y_d^{i1i4} \\ & \quad (3 S_4 y_C (g_1^2 + g_2^2) + S_Y (g_1^2 (-1 + 3 C_2 y_Y^2) + 3 g_2^2 (-1 + C_2 y_Y^2))) LF_{1,0}[m_e] + \\ & \quad \frac{1}{4} \frac{1}{m_e^2} S_Y^2 (3 S_Y^2 \overline{y_d}^{pR} \overline{y_d}^{i2i3} y_d^{p14} y_d^{i1r} + 3 \overline{y_d}^{p13} y_d^{i1i4} (S_Y^2 \overline{y_d}^{i2r} y_d^{pR} - C_Y^2 \overline{y_u}^{i2r} y_u^{pR}) - \\ & \quad \overline{y_d}^{i2i3} (2 y_d^{i1i4} (g_1^2 + 3 g_2^2) + 3 C_Y^2 y_d^{p14} \overline{y_u}^{pR} y_u^{i1r})) LF_{1,1}[m_e] + \\ & \quad \frac{1}{12} (S_Y^2 (3 \overline{y_d}^{i2i3} (y_d^{i1i4} (g_1^2 + 3 g_2^2) + 2 C_Y^2 y_d^{p14} \overline{y_u}^{pR} y_u^{i1r}) + 2 \overline{y_d}^{p13} (3 C_Y^2 y_d^{i1i4} \overline{y_u}^{i2r} y_u^{pR} - \\ & \quad 4 g_3^2 y_d^{p14} \delta_{i1i2})) - 4 g_3^2 (S_Y^2 \overline{y_d}^{i2p} y_d^{i1p} + C_Y^2 \overline{y_u}^{i2p} y_u^{i1p}) \delta_{i3i4}) LF_{1,2}[m_e] + \\ & \quad \frac{1}{9} g_1^2 \frac{1}{m_e^2} S_Y^2 \overline{y_d}^{i2i3} y_d^{i1i4} LF_{1,1,0}[m_1, m_d^{i3}] - \frac{1}{18} g_1^2 \frac{1}{m_e^2} S_Y^2 \overline{y_d}^{i2i3} y_d^{i1i4} LF_{2,1,-1}[m_1, m_d^{i3}] + \\ & \quad -\frac{1}{54} g_1^2 g_3^2 LF_{2,1,0}[m_1, m_d^{i3}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{54} g_1^2 g_3^2 LF_{2,2,-1}[m_1, m_d^{i3}] \delta_{i1i2} \delta_{i3i4} - \\ & \quad -\frac{1}{27} g_1^2 g_3^2 LF_{3,1,-1}[m_1, m_d^{i3}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{54} g_1^2 g_3^2 LF_{4,1,-2}[m_1, m_d^{i3}] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{1}{9} g_1^2 \frac{1}{m_e^2} S_Y^2 \overline{y_d}^{i2i3} y_d^{i1i4} LF_{1,1,0}[m_1, m_d^{i4}] - \frac{1}{18} g_1^2 \frac{1}{m_e^2} S_Y^2 \overline{y_d}^{i2i3} y_d^{i1i4} LF_{2,1,-1}[m_1, m_d^{i4}] + \\ & \quad -\frac{1}{54} g_1^2 g_3^2 LF_{2,1,0}[m_1, m_d^{i4}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{54} g_1^2 g_3^2 LF_{2,2,-1}[m_1, m_d^{i4}] \delta_{i1i2} \delta_{i3i4} - \\ & \quad -\frac{1}{27} g_1^2 g_3^2 LF_{3,1,-1}[m_1, m_d^{i4}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{54} g_1^2 g_3^2 LF_{4,1,-2}[m_1, m_d^{i4}] \delta_{i1i2} \delta_{i3i4} + \\ & \quad -\frac{1}{36} g_1^2 \frac{1}{m_e^2} S_Y^2 \overline{y_d}^{i2i3} y_d^{i1i4} LF_{1,1,0}[m_1, m_q^{i1}] - \\ & \quad -\frac{1}{72} g_1^2 \frac{1}{m_e^2} S_Y^2 \overline{y_d}^{i2i3} y_d^{i1i4} LF_{2,1,-1}[m_1, m_q^{i1}] + \frac{1}{216} g_1^2 g_3^2 LF_{2,1,0}[m_1, m_q^{i1}] \delta_{i1i2} \delta_{i3i4} + \\ & \quad -\frac{1}{216} g_1^2 g_3^2 LF_{2,2,-1}[m_1, m_q^{i1}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{108} g_1^2 g_3^2 LF_{3,1,-1}[m_1, m_q^{i1}] \delta_{i1i2} \delta_{i3i4} + \\ & \quad -\frac{1}{216} g_1^2 g_3^2 LF_{4,1,-2}[m_1, m_q^{i1}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{36} g_1^2 \frac{1}{m_e^2} S_Y^2 \overline{y_d}^{i2i3} y_d^{i1i4} LF_{1,1,0}[m_1, m_q^{i2}] - \\ & \quad -\frac{1}{72} g_1^2 \frac{1}{m_e^2} S_Y^2 \overline{y_d}^{i2i3} y_d^{i1i4} LF_{2,1,-1}[m_1, m_q^{i2}] + \frac{1}{216} g_1^2 g_3^2 LF_{2,1,0}[m_1, m_q^{i2}] \delta_{i1i2} \delta_{i3i4} + \\ & \quad -\frac{1}{216} g_1^2 g_3^2 LF_{2,2,-1}[m_1, m_q^{i2}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{108} g_1^2 g_3^2 LF_{3,1,-1}[m_1, m_q^{i2}] \delta_{i1i2} \delta_{i3i4} + \\ & \quad -\frac{1}{216} g_1^2 g_3^2 LF_{4,1,-2}[m_1, m_q^{i2}] \delta_{i1i2} \delta_{i3i4} + g_1^2 \frac{1}{m_e^4} S_Y^2 \overline{y_d}^{i2i3} y_d^{i1i4} LF_{1,1,-1}[m_1, \tilde{\mu}] + \\ & \quad 2 m_1 S_Y \tilde{\mu} C_Y g_1^2 \frac{1}{m_e^4} \overline{y_d}^{i2i3} y_d^{i1i4} (C_Y^2 - 2 S_Y^2) LF_{1,1,0}[m_1, \tilde{\mu}] + \\ & \quad -\frac{3}{4} g_2^2 \frac{1}{m_e^2} S_Y^2 \overline{y_d}^{i2i3} y_d^{i1i4} LF_{1,1,0}[m_2, m_q^{i1}] - \\ & \quad -\frac{3}{8} g_2^2 \frac{1}{m_e^2} S_Y^2 \overline{y_d}^{i2i3} y_d^{i1i4} LF_{2,1,-1}[m_2, m_q^{i1}] + \frac{1}{8$$